

BCA COLLEGE IDAR

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Applets in Java

Defination

- An **applet** is a small Java program that runs inside a web browser or applet viewer.
- Used mainly for **web-based applications** with graphical interface.

Types

1. **Local Applet** → stored on local system, loaded without a network.
2. **Remote Applet** → downloaded from a remote server via URL.

Applet Lifecycle

1. **init()** → called once, used for initialization.
2. **start()** → called every time the applet is started.
3. **stop()** → called when applet is stopped (e.g., user leaves the page).
4. **destroy()** → called before applet is destroyed.
5. **paint(Graphics g)** → used to draw graphics on applet window.

Creating and Running Applet

```
import java.applet.*;
import java.awt.*;
public class MyApplet extends Applet
{
    public void paint(Graphics g)
    {
        g.drawString("Hello Applet!", 50, 50);
    }
}
```

Run using appletviewer:

```
<applet code="MyApplet.class" width="300" height="200"></applet>
```

Passing Parameters

```
public void paint (Graphics g) {  
    String name = getParameter("userName");  
    g.drawString("Welcome " + name, 50, 50);  
}
```

In HTML (this code type in notepad and than save with .html file)

```
<applet code="MyApplet.class" width="200" height="100">  
    <param name="userName" value="Alnoor School">  
</applet>
```

3. Graphics Class in Java

- The `Graphics` class (in `java.awt`) is used for drawing shapes, text, and images in Java GUI or Applets.
- The `paint(Graphics g)` method is overridden to perform graphics operations.

Common Methods

1. `drawLine(int x1, int y1, int x2, int y2)` → draws a line.
2. `drawRect(int x, int y, int width, int height)` → draws rectangle.
3. `fillRect(int x, int y, int width, int height)` → fills rectangle.
4. `drawOval(int x, int y, int width, int height)` → draws oval.
5. `fillOval(int x, int y, int width, int height)` → fills oval.
6. `drawString(String str, int x, int y)` → draws text.
7. `setColor(Color c)` → sets color for drawing.
8. `drawPolygon(int[] xPoints, int[] yPoints, int n)` → draws polygon.

Example:

```
public void paint(Graphics g) {  
    g.setColor(Color.red);  
    g.drawLine(20, 30, 100, 30);  
    g.drawRect(50, 50, 150, 100);  
    g.fillOval(100, 100, 80, 80);  
    g.drawString("Graphics Demo", 60, 200);  
}
```